

Submission Report

1. Problem understanding

Environment: a grid warehouse with obstacles. Goal: move from S to G without crossing obstacles. Actions: Up, Down, Left, Right. The agent maintains its current state and search information. It is goal-based because it selects a path toward an explicit destination.

2. Agent design

Environment → current state → goal/search → next action. The implementation uses BFS to compute a collision-free path.

3. Prompt used

Write a well-documented Python goal-based agent for the supplied warehouse map. Represent the warehouse as a 2D grid, find a collision-free path from S to G, avoid obstacles, print the path or a no-path message, explain the chosen search algorithm, and test the program.

4. Execution result

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Algorithm: BFS goal-based path planning
Path found: True
Path length: 20
Path: [(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (2, 5), (2, 6), (2, 7), (1, 7), (1, 8), (1, 9), (1, 10), (1,
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5. Evaluation

The supplied map produced a valid path of length 20. A larger warehouse may require more memory and state expansion with BFS, motivating more scalable search strategies.